

MODULE-5

Electronic Waste (E-Waste) Management – Issues and Status in India and Globally, E-Waste Management Rules 2016 and Management Challenges

Introduction

Electronics waste, commonly known as e-scrap or e waste, is the trash generated from surplus, broken, and obsolete electronic devices. E-waste, encompasses discarded electrical and electronic equipment, including computers, phones, and appliances, and poses environmental and health risks due to the presence of toxic materials.

E-waste refers to all types of old, end-of-life, or discarded electrical and electronic equipment. This includes household appliances, office equipment, entertainment devices, lighting equipment, and tools. The amount of e-waste generated is increasing rapidly, driven by the rapid development of technology and a consumption-driven society. The sources and types of commonly generated e-waste is shown in figure 1 and 2 respectively.



Figure 1: Sources of E-waste



CATEGORIES OF E-WASTE



Figure 2: Types of E-waste.

E-Waste Generation Rate:

In 2022, a record 62 million tonnes of e-waste was produced, up 82% from 2010, and is projected to rise to 82 million tonnes by 2030. Only 22.3% of the generated e-waste was documented as formally collected and recycled in an environmentally sound manner. The global average is 7.8 kg of e-waste per capita per year. The generation of electronic waste is rising almost five times faster than documented e-waste recycling. India is the 3rd largest e-waste generator globally, with 4.1 billion kg generated in 2022. The total E-waste in India has been estimated to be 1,86,180 Tonnes per year.

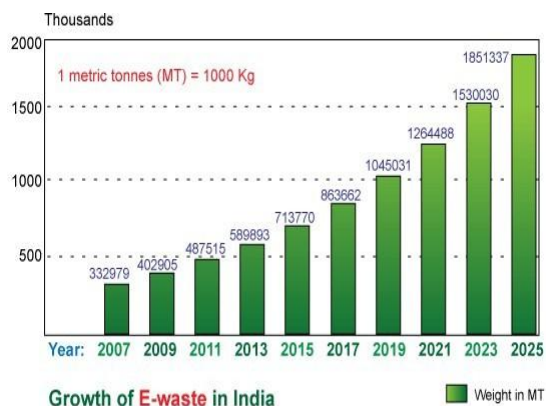
E-Waste Growth and Per capita Analysis-Globally: E-waste is a health and environmental hazard, containing toxic additives and hazardous substances.

- UNU report shows steady growth in e-waste, projected to reach nearly 50 million metric tonnes by 2018.
- About one quarter of total e-waste is personal digital devices, with household appliances and heating/cooling equipment accounting for the rest.
- Asia generated the most e-waste in 2014, with 16 million metric tonnes.
- Europe generates the most e-waste per capita, with 15.6 kg per inhabitant.
- A record 62 million tonnes of e-waste was produced in 2022, up 82% from 2010.
- The report predicts a drop in the documented collection and recycling rate from 22.3% in 2022 to 20% by 2030.
- Challenges include technological progress, higher consumption, limited repair options, shorter product life cycles, growing electrification, design shortcomings, and inadequate e-waste management infrastructure.
- If countries could increase e-waste collection and recycling rates to 60% by 2030, benefits would exceed costs by over \$38 billion.
- Less than 23.2% of the year's e-waste mass was properly collected and recycled in 2022, leaving US\$ 62 billion worth of recoverable natural resources unaccounted for.

The annual generation of e-waste is rising by 2.6 million tonnes annually, on track to reach 82 million tonnes by 2030.

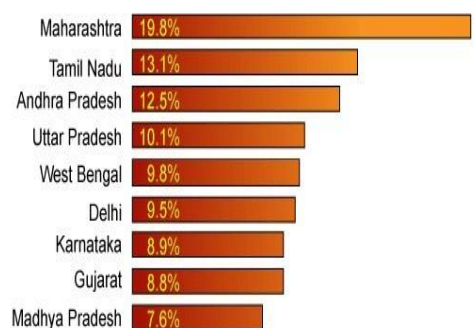
E-waste production in India:

- India is the second largest electronic waste generator in Asia, with an expected 20% annual increase.
- The country has two types of electronic waste: organized (10%) and unorganized (90%).
- 90% of the waste is in the unorganized market, contributing to 70% of overall toxic wastes in landfills.
- The Indian PC industry is growing at a rate of 25% annually as shown in figure 4.
- Only 40% of the total electronic waste is recycled, with 60% remaining in warehouses due to inefficient collection systems.
- E-waste scrap is managed through reuse, back yard recycling, and municipal dumping.
- The top ten cities generating e-waste as shown in figure 5 are Mumbai, followed by Tamil Nadu, Andhra Pradesh, Uttar Pradesh, West Bengal, Delhi, Karnataka, Gujarat, Madhya Pradesh, and Punjab.



Source: Department of Information Technology

Chart: CopperBridge Media



State-wise E-waste Generation in India (Tonnes/year)

Source: Department of Information Technology

Chart: CopperBridge Media

Fig 4. Projection of E-waste generation in India.
Fig 5. State wise contribution to E-waste in India.

The Central Pollution Control Board (CPCB) estimates the e-waste generation at national level based on the countrywide sales data provided by producers and average life of notified electrical and electronic equipment (EEE), as mandated under the E-waste Management Rules, 2016. As per the information available with CPCB, e-waste generated in the country from twenty-one (21) types of EEE notified under the E-Waste (Management) Rules, 2016 since Financial Year (FY) 2017-18 is given below:

Financial Year	Generation (Tonnes)
2017-18	7,08,445.00
2018-19	7,71,215.00
2019-20	10,14,961.21
2020-21	13,46,496.31
2021-22	16,01,155.36

The quantity of WEEE generated in India is illustrated in figure 3.

Quantity of WEEE (Waste Electrical and Electr Equipment) generated in Indian States. .

SN	States	WEEE (Tonnes)	SN	States	WEEE (Tonnes)
1	Andaman and Nicobar Islands	92.2	19	Lakshadweep	7.4
2	Andra Pradesh	12,780.3	20	Madhya Pradesh	7,800.6
3	Arunachal Pradesh	131.7	21	Maharashtra	20,270.6
4	Assam	2,176.7	22	Manipur	231.7
5	Bihar	3,055.6	23	Meghalaya	211.6
6	Chandigarh	359.7	24	Mizoram	79.6
7	Chhattisgarh	2,149.9	25	Nagaland	145.1
8	Dadra and Nagar Haveli	29.4	26	Orissa	2,937.8
9	Daman and Diu	40.8	27	Puducherry	284.2
10	Delhi	9,729.2	28	Punjab	6,958.5
11	Goa	427.4	29	Rajasthan	6,326.9
12	Gujarat	8,994.3	30	Sikkim	78.1
13	Haryana	4,506.9	31	Tamil Nadu	13,486.2
14	Himachal Pradesh	1,595.1	32	Tripura	378.3
15	Jammu and Kashmir	1,521.5	33	Uttar Pradesh	10,381.1
16	Jharkhand	2,021.6	34	Uttarakhand	1,641.1
17	Karnataka	9,118.7	35	West Bengal	10,059.4
18	Kerala	6,171.8		Total	146,180.7

Country Level WEEE assessment study by IRGSSA (2005)

Fig 3: Quantity of WEEE generated in India

Composition of E-waste

E-waste composition typically includes metals (like gold, silver, and copper), plastics, glass, and toxic materials like lead, mercury, and cadmium. Figure 6 illustrates the distinctive materials in a WEEE.

- **Metals:** E-waste contains various metals, including gold, silver, copper, and less common metals like platinum, palladium, and rhodium.
- **Plastics:** A significant portion of e-waste is made up of different types of plastics, including PVC and flame-retardant materials.
- **Glass:** Screens, lenses, and other components in electronic devices contain glass, which is also a part of e-waste.
- **Toxic Materials:** E-waste can contain hazardous substances like lead, mercury, cadmium, and beryllium, which pose risks to human health and the environment.
- **Other materials:** E-waste also contains metal oxides, ceramics, polyaromatic hydrocarbons (PAH), PCBs, PBDEs, and polyvinyl chloride phthalates.

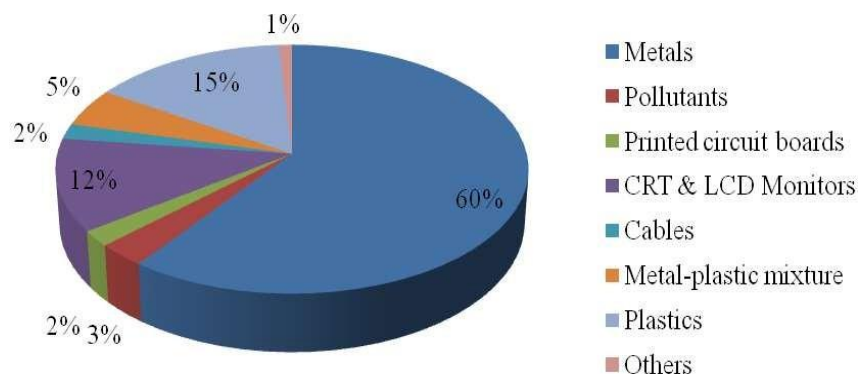


Fig 6: The distinctive contents of a WEEE.

Environmental and Health Risks:

E-waste contains hazardous substances like lead, mercury, cadmium, and PVC, which can contaminate soil, water, and air. Exposure to these toxins can cause serious health problems, including nervous system damage and respiratory issues as shown in table 1.

Proper recycling of e-waste is crucial to prevent these materials from entering the environment and causing harm. Some e-waste is illegally dumped, leading to further environmental contamination and health risks.

Table 1: Effects of E-waste constituent on health

Source of e-wastes	Constituent	Health effects
Solder in printed circuit boards, glass panels and gaskets in computer monitors	Lead (PB)	- Damage to central and peripheral nervous systems, blood systems and kidney damage. - Affects brain development of children.
Chip resistors and semiconductors	Cadmium (CD)	- Toxic irreversible effects on human health. - Accumulates in kidney and liver. - Causes neural damage. - Teratogenic.
Relays and switches, printed circuit boards	Mercury (Hg)	- Chronic damage to the brain. - Respiratory and skin disorders due to bioaccumulation in fishes.
Corrosion protection of untreated and galvanized steel plates, decorator or hardner for steel housings	Hexavalent chromium (Cr) VI	- Asthmatic bronchitis. - DNA damage.
Cabling and computer housing	Plastics including PVC	Burning produces dioxin. It causes - Reproductive and developmental problems; - Immune system damage; - Interefere with regulatory hormones
Plastic housing of electronic equipments and circuit boards.	Brominated flame retardants (BFR)	Disrupts endocrine system functions
Front panel of CRTs	Barium (Ba)	Short term exposure causes: - Muscle weakness; - Damage to heart, liver and spleen.
Motherboard	Beryllium (Be)	- Carcinogenic (lung cancer) - Inhalation of fumes and dust. Causes chronic beryllium disease or berylliosis. - Skin diseases such as warts.

Some non-hazardous materials found in e-waste are presented in Table 2:

Table 2. Non-Hazardous substances found in E-waste and their sources.

Non-Hazardous Substances	Source
Tin	Solder, coatings on component leads.
Copper	Copper wire, printed circuit boards tracks, component leads.
Aluminium	Nearly all electronic goods using more than a few watts of power (heat sinks),electrolytic capacitors.
Iron	Steel chassis, cases, and fixings.
Germanium	Transistorized electronics (bipolar junction transistors).
Silicon	Diodes, Transistors and IC's.
Zinc	Plating.

What is E-Waste Management?

E-waste management is a process to collect e-waste, recover and recycle material by safe methods, dispose of e-waste by suitable techniques to reduce its adverse impacts on environment. E-waste management refers to handling and managing electronic waste, such as phones, computers, tablets, and more.

Challenges of Managing E-Waste

There are quite a few challenges to managing e-waste that make it hard for organizations to practice it. While these are difficult challenges, organizations must overcome them to establish an effective and efficient workflow for managing e-waste. Here are some of the key challenges in e-waste management that companies today face:

- Ignorance of the importance of practicing proper e-waste management within an organization
- Poor infrastructure for managing e-waste in certain areas
- Short lifespan of modern electronics
- Not enough regulations and enforcement by local authorities.
- Technological advances speed up obsolescence & lead to more e- waste.

E-Waste Management and Toxic Substances

- Electrical and electronic items contain toxic substances, which can be released into the environment when managed improperly.
- Unsound practices at e-waste sites include scavenging, dumping, landfilling, open burning, acid baths, plastic shredding, and manual disassembly.
- These activities release toxic pollutants, contaminating air, soil, dust, and water.
- Open burning and heating are the most hazardous due to the toxic fumes they create.
- These pollutants can travel significant distances, exposing people in faraway areas to hazardous substances.

E-Waste Recycling and Adverse Health Outcomes

- Children and pregnant women are particularly vulnerable to hazardous pollutants from informal e-waste recycling activities.

- Children often engage in waste picking, scavenging, burning, and manual dismantlement, which exposes them to injury and high levels of hazardous substances.
- E-waste exposure may lead to adverse neonatal outcomes, neurodevelopment, learning, and behaviour outcomes, especially associated with lead.
- Reduced lung and respiratory function and increased asthma incidence may be linked to high levels of contaminated air pollution from e-waste recycling sites.
- Children and pregnant women are at high risk due to their unique vulnerabilities, with e-waste recycling activities releasing toxic chemicals that can contaminate breastmilk.
- E-waste contains neurotoxins like lead and mercury, which may disrupt the development of the central nervous system during pregnancy, infancy, childhood, and adolescence.

Prevention and Management of E-Waste Recycling: The toxins contribute to e-waste's slow-burn problem for health and the planet. Proper recycling can help control these toxins, but too much ends up dumped or burned.

- Adopt and enforce high-level international agreements.
- Develop and implement national e-waste management legislation.
- Incorporate health protection measures into national legislation.
- Monitor e-waste sites and communities.
- Implement interventions to improve informal e-waste recycling activities.
- Educate health workers on e-waste-related child health issues.
- Eliminate child labour.

International Agreements

- Basel Convention controls transboundary movement of hazardous wastes.
- Ban Amendment to Basel Convention in 1998 prohibits movement of hazardous wastes.
- Runs programmes and workshops on environmentally sound e-waste management.
- Regional conventions like Bamako Convention and Waigani Convention restrict e-waste movement in African and South Pacific countries.

E-Waste management in India:

India is a major e-waste producer, ranking among the top five globally after the United States, China, Japan, and Germany. The majority of e-waste in India is managed by the informal sector, with over 95% of e-waste being illegally recycled by informal waste pickers. India lacks adequate infrastructure for formal e-waste collection and recycling, leading to improper disposal and environmental contamination.

The informal recycling practices in India, often involving open burning and dismantling, lead to the release of toxic substances into the environment, affecting air, water, and soil quality, and posing health risks to workers and nearby communities.

India generates over 2 million tonnes of e-waste annually and also imports massive volumes of e-waste from other countries. There is a lack of public awareness about the hazards of e-waste and the penalties for improper disposal, and consumers lack financial incentives for responsible disposal.

Need for a Multi-Pronged Approach:

Addressing India's e-waste problem requires a multi-pronged approach, including strengthening regulations, promoting responsible consumption, investing in infrastructure, and engaging the informal sector.

E-waste management in India is predominantly informal, with approximately 90% of e-waste collection and 70% of recycling being managed by a competitive informal sector. The volume of e-waste in India has witnessed a significant surge to 1.6 million tonnes in 2021-22. Over 2 Million old PCs ready for Disposal in India State wise contribution of E- waste in India

E-Waste Recycling Challenges in India

- Inefficient and hazardous processing of e-waste due to leakages from formal to informal sectors.
- Many e-waste recyclers in India are not recycling waste and are storing it in hazardous conditions.
- The informal e-waste trade chain is characterized by an entrepreneurial SME infrastructure, allowing profitability in operations.
- Unauthorized facilities and informal recyclers use unsafe waste management systems and sub-standard technologies.
- High collection rates for various categories of e-waste, including high-value WEEE like computers, laptops, televisions, and mobiles.
- Dismantling and recycling operations are crude and expose individuals to hazardous chemicals and environmental damages.
- E-waste contains potent toxins like lead, cadmium, and mercury, which can be hazardous to workers when not processed cautiously.

E-waste Management Rules in India

The Environment, Forest, and Climate Change Ministry (MoEF&CC) have announced the E-Waste Management Rules 2016. These new rules replaced the earlier E-Waste (Management and Handling) Rules of 2011.

Introduction

- Increasing use and dependence on Electrical and Electronic Equipment (EEE), such as, Mobile Phones, Personal Computers, Laptops, Servers, Data Storage Devices, Photo-copying Machines, Televisions, Washing Machines, Refrigerators, Air Conditioners, etc. resulting into generation of large quantities of e-Waste.
- The EEE have valuable materials and hazardous/ toxics substances in their components. After their useful life, they may not cause any harm if stored safely in households/stores, however, if the e-Waste is opened-up and attempts are made for retrieval of useful components or material in an un-scientific manner or if the material is disposed in open, then it may cause health risks and damage to environment.

Objective & Application

To enable the recovery and/or reuse of useful material from Waste Electrical and Electronic Equipment (WEEE), thereby reducing the hazardous wastes destined for disposal and to ensure environmentally sound management of all types of WEEE. These rules may be called the E-Waste (Management) Rules, 2016.

Application

Apply to every manufacturer, producer, consumer, bulk consumer, collection centres, dealers, e-retailer, refurbisher, dismantler and recycler involved in manufacture, sale, transfer, purchase, collection, storage and processing of e-waste or electrical and electronic equipment listed in **Schedule**

I, including their components, consumables, parts and spares which make the product operational but shall not apply to –

- (a) Used lead acid batteries as covered under the Batteries (Management and Handling) Rules, 2001 made under the Act.
- (b) Micro enterprises as defined in the Micro, Small and Medium Enterprises Development Act, 2006 (27 of 2006); and 2.
- (c) Radio-active wastes as covered under the provisions of the Atomic Energy Act, 1962 (33 of 1962) and rules made there under.

Responsibilities of the manufacturer

- ☐ Collect e-waste generated during the manufacture and channelise it for recycling or disposal.
- ☐ Apply for an authorization in Form 1 (a) from the concerned State Pollution Control Board.
- ☐ Ensure that no damage is caused to the environment during storage and transportation of e-waste.
- ☐ Maintain records of the e-waste generated, handled and disposed.
- ☐ File annual returns in Form-3, to the concerned State Pollution Control Board on or before the 30th day of June following the financial year to which that return relates.

Responsibilities of the producer

Implementing the Extended Producers Responsibility with the following frameworks, namely:

- (a) Collection and channelization of e-waste generated from the 'end-of-life' products with the targets prescribed in Sch. III in EPR – Authorizations.
- (b) The mechanism used for authorized dismantler or recycler shall be as per EPR - authorization.
- (c) For disposal in Treatment, Storage and Disposal Facility, a pre-treatment is necessary to immobilize the mercury and reduce the volume of waste.
- (d) EPR should comprise of general scheme for collection of e-waste from the market placed earlier, such as through dealer, collection centers, Producer Responsibilities organization, through buy-back arrangement, exchange scheme, deposit refund system etc. and channelizing to authorized recyclers.
- (e) Providing contact details such as address, e-mail address, toll-free telephone numbers or helpline numbers to consumer(s) or bulk consumer(s) through their website and product user documentation so as to facilitate return of end-of-life electrical and electronic equipment.
- (f) Creating awareness through media, publications, advertisements, posters, or by any other means of communication.

Responsibilities of consumer or bulk consumer

- Consumers or bulk consumers shall ensure that e-waste generated by them is channelized through collection center or dealer of authorized producer or dismantler or recycler or through the designated take back service provider of the producer to authorized dismantler or recycler.
- Bulk consumers shall maintain record of e-waste generated by them in Form-2 and make such records available for scrutiny by the concerned State Pollution Control Board.
- Consumers or bulk consumers shall ensure that such end-of-life electrical and electronic equipment are not admixed with e-waste containing radioactive material as covered under the provisions of the Atomic Energy Act, 1962 (33 of 1962) and rules made there under.
- Bulk consumers shall file annual returns in Form-3, to the concerned State Pollution Control Board on or before the 30th day of June following the financial year to which that return relates. In case

of the bulk consumer with multiple offices in a State, one annual return combining information from all the offices shall be filed to the concerned State Pollution Control Board on or before the 30th day of June following the financial year to which that return relates.

Storage of e-waste

- Every manufacturer, producer, bulk consumer, collection center, dealer, refurbisher, dismantler and recycler may store the e-waste for a period not exceeding one hundred and eighty days and
- Shall maintain a record of collection, sale, transfer, and storage of wastes and make these records available for inspection:

Provided that the concerned State Pollution Control Board may extend the said period up to three hundred and sixty-five days in case the waste needs to be specifically stored for development of a process for its recycling or reuse.

India's Environmental Regulation and Waste Management

- The Environment (Protection) Act of 1986 focuses on preventing, controlling, and abatement of environmental pollutants.
- The E-Waste (Management and Handling) Rules, 2011 regulate e-waste management, involving stakeholders from EEE producers to recyclers.
- The 2016 Amendment added more stakeholders, including PROs, and outlined their roles and responsibilities.
- The 2018 Amendment revised collection targets and mandated PROs to register with the CPCB for monitoring and auditing.
- The 2022 MoEF&CC notified the E-waste Management Rules, limiting the rules to four stakeholders: producers, manufacturers, recyclers, and refurbishers.
- The rules define responsibilities of various bodies, including CPCB/SPCBs, manufacturers, recyclers, state government, urban local bodies, and port authorities.

Legislations on E-Waste

As per E-Waste (Management) Rules, 2022, Rule 17, Schedule - V the responsibilities of Urban Local Bodies are defined as follows:

- (1) To ensure that e-waste if found to be mixed with Municipal Solid Waste is properly segregated, collected and is channelised to registered recycler or refurbisher.
- (2) To ensure that e-waste pertaining to orphan products is collected and channelized to registered recycler or refurbisher.
- (3) To facilitate setting up e-waste collection, segregation, and disposal systems.
- (4) Conducting training sessions to develop capacities of the urban and rural local bodies.

E-waste Management: E-waste management should begin at the point of generation. This can be done by:-

- Waste minimization techniques by sustainable product design.
- Regulatory Compliance and Follow Agreed Standards.

Steps for E-waste management: To use things with care to reduce the amount of waste generated. To repeat use of items or parts of items. To use waste as resources.

- Step 1: Use 3R technique to minimize the E-waste.
- Step 2: Identify the E-waste category item.
- Step 3: Identify the E-waste composition or determine it.
- Step 4: Identify possible hazardous content in E- waste.
- Step 5: Identify, whether the E-waste component is hazardous or the entire E-waste item is hazardous.

The e-waste recycling process: Electronics recycling is the process of recovering material from old devices to use in new products. Recycling e-waste involves several steps, each designed to handle and recover valuable materials from old electronic devices safely, which is explained in simple terms as shown in fig 7.



Fig 7: Steps in recycling E-waste.

Step 1: Collection-The first step is gathering old electronics through designated recycling bins, drop-off centres, or take-back programs. Local governments, businesses, or community events may run these collection points. Some services even offer on-demand pickups to make recycling easier. In this step proper separation rules. Many centres use bins for various types of e-waste, such as batteries, phones, or computers, to prevent harmful substances from mixing. Batteries, for instance, require special care because they can cause fires or leak toxic chemicals.

Step 2: Safe Storage- Once collected, e-waste is stored safely until it's ready for processing. This is especially important for items like old TVs and monitor screens containing lead, which can harm people and the environment if mishandled. In the past, glass from these screens was recycled into new products, but declining demand has left much of it in storage.

Step 3: Sorting, Dismantling, and Shredding-Next, the e-waste goes through manual sorting. Workers remove specific items that need special processing, such as batteries or lightbulbs. Some parts, like reusable components or valuable metals, are separated at this stage. After sorting, the remaining e-waste is shredded into tiny pieces, usually just a few centimetres in size. This makes separating materials like metals, plastics, and glass more accessible.

Step 4: Mechanical Separation-Uses advanced machines to sort materials based on their properties: Magnetic Separation-Large magnets pull out metals like steel and iron. Nonferrous metals like aluminum and copper are separated using eddy currents. Water Separation-Plastic and glass pieces are separated using water. Heavy materials sink, while lighter ones float, making it easier to sort and purify each type. These processes ensure that every material is collected and prepared for the next stage.

Step 5: Material Recovery-Finally, the separated materials are cleaned and prepared for reuse. Metals like steel and copper may be used in smelting plants to create new products. Plastics might be melted down and turned into new items, while reusable components can go directly back into production. This stage is vital for creating a circular economy. By recovering and reusing materials, the need for mining raw resources reduces, which helps protect the environment and save energy.

Benefit of recycling of E-waste:

- E-waste contains many valuable, recoverable materials such as aluminium, copper, gold, silver, plastics, and ferrous metals. In order to conserve natural resources and the energy needed to produce new electronic equipment from virgin resources, electronic equipment can be refurbished, reused, and recycled instead of being landfilled.
- E-waste also contains toxic and hazardous materials including mercury, lead, cadmium, beryllium, chromium, and chemical flame retardants, which have the potential to leach into our soil and water.
- Protects your surroundings- Safe recycling of outdated electronics promotes sound management of toxic chemicals such as lead and mercury.
- Conserves natural resources- Recycling recovers valuable materials from old electronics that can be used to make new products. As a result, we save energy, reduce pollution, reduce greenhouse gas emissions, and save resources by extracting fewer raw materials from the earth.
- Helps others- Donating your used electronics benefits your community by passing on ready-to-use or refurbished equipment to those who need it.
- Saves landfill space - E-waste is a growing waste stream. By recycling these items, landfill space is conserved.

E-Waste Management for Cities

E-waste management should be considered in the broader sustainability context and not in isolation. E-waste management, recycling, refurbishing, and circular economy in manufacturing (e.g., using recycled materials in manufacturing) are all interconnected sustainability practices. The smart cities should align the policies and approaches targeted at promoting e-waste recycling and sync with those formulated to promote a circular economy in electronics and other sectors.

The life cycle of any electronic product may be considered in three phases: pre-consumption, consumption, and post-consumption.

Pre-consumption	Consumption	Post-consumption
Mining		E-waste management
Material Extraction Design Manufacturing sales andDistribution	Use Servicing Warranty Repair	Collection and Disassembly Material recovery Refurbishment

Post-Consumption Recycling and Its Interconnectedness:

- Recycling is linked to activities under the 'Pre-consumption', 'Consumption', and 'Post-consumption' phases.
- Sustainable design in the 'Pre-consumption' phase minimizes waste production, increasing ease

and scope of recycling in the 'Post-consumption' phase.

- Sustainable design in the 'Pre-consumption' phase facilitates easier dismantling of electronic products and higher material recovery rates from collected e-waste.
- There's a disconnect between the 'Pre-consumption', 'Consumption', and 'Post-consumption' phases for most electronic products.
- Rapid technological advancements in electronic product manufacturing have led to changing material compositions.
- There's a lack of communication between designers and manufacturers and refurbishers and recyclers.
- Unchecked e-waste volume and underlying technology complexity make it difficult to manage and recycle.

Smart Cities' E-Waste Management Strategy

- Establish a database of electronic repair shops in smart cities.
- These shops should maintain a material flow accounting of e-waste and e-waste components.
- The database will help prepare an inventory of e-waste, facilitating compliance with regulations and attracting recyclers.
- Convert current informal sector hubs to 'Micro Enterprise Clusters', with in-situ development as the strategy.
- Encourage individuals and entities in the informal sector to own shops or connect with others in electronic hubs.
- Organise periodic information dissemination, training sessions, and workshops for the informal sector entrepreneurs and workers.
- Engage other key stakeholders with the informal sector and facilitate cross-sector partnerships.
- Use existing IT infrastructure to create a knowledge bank of systems, practices, and approaches to manage e-waste.
- Employ a multi-pronged approach for public engagement with e-waste management and recycling.
- Awareness campaigns and monthly collection drives at schools, colleges, and commercial establishments can help educate the public about e-waste management and disposal options.

Stakeholder Engagement and EPR Framework for Smart City

- Insufficient engagement of producers with key stakeholders: informal sector, formal recyclers, and local governments.
- EPR framework entrusts different e-waste flow responsibilities to these stakeholders.
- Producers can be incentivised to engage with smart city's recyclers and administrators.
- Expected introduction of 'right to repair' rule similar to EU's.
- Database of electronic repair shops, physical electronic hubs, integrated informal sector, and knowledge bank can facilitate producer involvement.